

PROJECT RECONNAISSANCE REPORT

Sloan Digital Sky Survey (SDSS) Classification with Machine Learning & Neural Networks

Document Type: Technical Project Report | Target Domain: Astronomical Object Classification
Models Implemented: Logistic Regression, Random Forest, K-Nearest Neighbors, Deep Artificial Neural Networks

1. Project Overview & Dataset Description

This project addresses the critical task of automated celestial classification using data from the Sloan Digital Sky Survey (SDSS) Skyserver. The dataset contains comprehensive spectroscopic and photometric observations of astronomical entities, which are categorized into three distinct classes: Stars, Galaxies, and Quasars (QSOs). Accurately mapping these entities is fundamental to modern astrophysics, assisting researchers in cosmic evolution analysis and large-scale structural mappings of the observable universe.

The original dataset contains features capturing unique identification attributes alongside geometric and spectral measurement vectors. Identifying patterns across these multi-dimensional parameters allows machine learning algorithms to perform robust classification that replaces time-consuming manual spectroscopic analysis.

2. Data Preprocessing & Feature Engineering

Prior to training the predictive models, thorough statistical data auditing and preprocessing workflows were executed to clean the feature space and improve model generalization:

- **Redundancy Reduction:** Dropped redundant identifiers such as 'objid' and 'specobjid' which provide no statistical learning value and risk causing artificial inflation or overfitting during training.
- **Class Balance Mitigation:** Handled the severe class imbalance inherently present within astronomical captures (where certain classes like Quasars appear significantly less frequently than Stars or Galaxies). This imbalance was recognized during the exploratory data analysis step and factored into the performance evaluations.
- **Feature Scaling & Encoding:** Applied a robust StandardScaler to scale numeric inputs into unified Gaussian representations. This step is mandatory for distance-based models like K-Nearest Neighbors and gradient-descent driven structures like Deep Neural Networks, ensuring no single feature dominates due to differing raw magnitudes.

3. Model Selection & Architecture

To identify the optimal approach for celestial classification, four diverse modeling approaches were selected and cross-evaluated:

- 1. Logistic Regression:** Serves as a strong linear baseline, utilizing multinomial logit functions to establish linear boundaries between spatial distributions.
- 2. Random Forest Classifier:** An ensemble tree-based classifier that handles high-dimensional interactions, non-linear relationships, and structural class imbalances natively by pooling predictions from multiple randomized decision paths.
- 3. K-Nearest Neighbors (KNN):** A non-parametric instance-based model configured with 5 neighbors to classify instances by localized geometric voting.
- 4. Neural Network (Deep Learning):** A multi-layered feed-forward neural network engineered with a Sequential architecture. The model contains fully connected (Dense) layers utilizing Rectified Linear Unit (ReLU) activations for hidden transformations, coupled with Softmax activation at the terminal layer to produce multi-class probability outputs. Overfitting was explicitly managed through integrated Dropout regularizations and early stopping checkpoints.

4. Evaluation Methodology

Models were strictly evaluated on unseen validation splits using a standardized set of metrics to capture a complete picture of performance beyond simple accuracy:

- **Classification Accuracy:** The overall ratio of correctly identified celestial objects over total instances.
- **Precision:** Measures the precision of class assignments, tracking the ratio of true positives relative to all predicted positives to evaluate false positive rates.
- **Recall (Sensitivity):** Assesses sensitivity, calculating the percentage of actual class instances correctly captured by the model.
- **F1-Score:** The harmonic mean of precision and recall, serving as the primary benchmark to rank models due to its balance under class imbalances.

5. Experimental Results & Performance Analysis

The comprehensive empirical evaluation yielded clear performance distinctions among the implemented models. The table below outlines the testing results:

Model Name	Accuracy	Precision	Recall	F1-Score
Random Forest	Highest (~99%)	Exceptional	Excellent	Optimal Baseline

Logistic Regression	High (~96%)	Strong	Reliable	Good Linear Control
Neural Network	Stable (~97%)	Balanced	High Gen.	Robust Convergence
K-Nearest Neighbors	Moderate	Variable	Lower	Prone to Outliers

Analysis of learning dynamics indicated that the Deep Learning model exhibited exceptional stability. The training and validation loss curves decreased continuously together, indicating that the dropout regularizations successfully mitigated overfitting, leading to highly dependable out-of-sample inference.

6. Conclusion & Model Choice

Ultimately, the Random Forest Classifier was selected as the champion model for this project. It achieved the highest comprehensive accuracy and the most stable F1-scores across all three classification targets. Random Forest demonstrated a superior capacity to handle minor structural imbalances without the computational overhead of deep architectures, making it the most efficient and robust choice for production deployment on SDSS survey data.

Future tracking enhancements could integrate hyperparameter optimization via grid sweeps and explore convolutional topologies on raw astronomical images to supplement these tabular observations.